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Do Consumers Value Diversity in Agriculture? A Choice Experiment

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ABSTRACT

Farm ownership in the US today is racially and ethnically homogeneous. Diverse farmers and ranchers make up less than 10% of all producers, they tend to be less financially stable and have tighter profit margins. In the past few years, both government and industry programs have aimed to increase the participation and profitability of diverse farmers, some of which have been rolled back in response to recent legal and political changes. Despite active debate, little is known about public support for diversity in agriculture. Here, we explore consumer demand for diversity in US agriculture by measuring willingness to pay for a diverse farmer label. We find evidence of a positive price premium for a diverse farmer label on average, with significant consumer heterogeneity. Demand differed across a variety of consumer attributes, including consumers' politics, gender, and connection to agriculture. Our results provide timely insights on public demand for diversity in agriculture.

JEL Classification: Q18, D10, D12, D9, M31

1 | Introduction

Farmers from racially or ethnically diverse backgrounds, sometimes termed Socially Disadvantaged Farmers and Ranchers (SDFRs), represent less than 10% of all US producers (US Department of Agriculture 2020).¹ The participation of SDFRs, and consequentially their financial outcomes, has a complex history (Casey 2021; Hutchins and De Marinis 2025). Research has shown that racial and ethnic minorities have been historically excluded from land ownership (Muhammad et al. 2024), Extension education (Harris 2008), and federal support programs (Teal and Stevens 2024; Hendricks et al. 2024; Yu and Lim 2024; US Commission on Civil Rights 1965; Cowan and Feder 2013; USDA Equity Commission 2023). Today, SDFRs are generally smaller in scale, less likely to receive government payments, and exhibit lower financial stability (Todd et al. 2024).

In recent years, there have been efforts to increase diversity in business ownership, including programs aimed at businesses operating in the food system. These efforts included investments and commitments by major food retailers (e.g., Walmart/Sam's Club, Target, Kroger, Albertsons, Meijer, Schnucks, Giant Eagle, Trader Joe's, etc.) and food brands (e.g., Kellogg's) to increase the diversity of suppliers (Browne 2021; Goldschmidt 2021; Grace 2020; Hurtig 2020; International Supermarket News 2024; Martin 2020; Moran 2022; Strailey 2021a, 2021b). Diverse supplier labeling initiatives have also been introduced for restaurants. For example, both Yelp and Google Maps allow owners to identify race, ethnicity, gender, and other attributes (Brown 2020; Lyles 2020). Similarly, federal efforts have also been made. For instance, the USDA implemented several policies aimed at increasing SDFR profitability or participation through increasing access to credit or Extension education

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programs (e.g., USDA's Program 2501, targeted Farm Service Agency loans) (Johnson 2023; Mishra et al. 2024; Vekemans et al. 2024). Yet, some of these actions have been recently rolled back by the government and private firms in response to changes in the legal and political environment (Holman and Creswell 2025; The White House 2025a). For example, some firms have scaled back diversity, equity, and inclusion (DEI) policies, and the USDA has removed grant support for programs that included DEI activities (Held 2025; The White House 2025a; USDA 2025a; USDAa 2025b).²

Despite these programs and active debate for diverse business ownership broadly and within some areas of the food system, little is known about public demand for diversity in agriculture. In this paper, we explore consumer demand for diversity in US agriculture by measuring willingness to pay for a diverse farmer label. Specifically, we utilize an online discrete choice experiment to evaluate consumer demand for a hypothetical label that identifies products from farms owned by racially or ethnically diverse farmers.

Results from our mixed logit model show that, on average, consumers have a positive valuation for diversity in agriculture, with the label commanding an average price premium of approximately 5%. Importantly, demand for diversity in agriculture differed substantially across consumers. Using individual estimates of willingness to pay for a diverse farmer label, we find that liberal consumers and female consumers displayed stronger preferences for diversity in agriculture. To further explore heterogeneity, we utilize a latent class model to understand how demand differed across consumer segments. We show that one class of consumers (which accounted for about 36% of the sample) had strong demand for diversity in agriculture, finding a 21% premium for the diversity label; whereas the other consumer segments did not. Consumers in the class with positive preferences for diversity were more likely to be female consumers, younger consumers, and those without personal or familial connections to agriculture.

We begin by discussing the relevant literature, highlighting the role of food labels, heterogeneous responses to labels, and demand for diversity. Second, we provide details about our experimental design, survey design, and data before discussing our methods of demand estimation. Finally, we discuss our results and provide implications for policymakers and other stakeholders.

2 | Literature Review

When consumers make choices, some attributes—search attributes—can be evaluated prior to purchase, whereas others—credence attributes—cannot. In the case of credence attributes, labels can reduce information asymmetry, allowing consumers to make choices that better align with their true preferences (Caswell and Anders 2011). For example, a consumer purchasing a food product can see and assess some aspects of the product (e.g., size, color), but without clear labeling, cannot assess a variety of credence attributes (e.g., organic, cage-free). Understanding consumer demand for particular labels is also

useful, as consumers' preferences play a fundamental role in shaping the direction of food production. For example, the food system has witnessed the influence of consumer preferences on production decisions many times, with attributes such as “organic” and “locally grown” being recent examples.

While consumers' preferences are mostly driven by attributes that directly benefit them, such as taste, safety, and convenience, consumers are also often willing to pay for credence attributes for more altruistic reasons (Caswell and Anders 2011). This phenomenon has been studied more extensively in relation to labels that highlight environmental benefits. The distinction between benefits felt directly by the consumer and altruistic ones is sometimes formalized, for example, by separating utility into two parts (material and expressive utility) or by analyzing the non-material benefits associated with the “warm glow” effect (i.e., positive feelings for doing a good thing) (Bronnmann et al. 2021; Hartmann and Apaolaza-Ibáñez 2012; Kalaitzandonakes et al. 2024; Nurse Rainbolt et al. 2012; Tsai 2005). Labels, however, can impact both. For example, a local label might increase a consumer's utility if they expect the produce to be tastier (e.g., having been picked when ripe rather than early to facilitate transport). A local label might also increase a consumer's utility if they feel good about their decision to support farmers in their own community.

Previous research has shown that consumers are willing to pay premiums to support farms with attributes they care about. As with other credence attributes, farm ownership information is not easily assessed by consumers without labels.³ So far, this research has focused heavily on the geographic attributes of a farm, including local, regional, and national labeling programs (Thøgersen 2023) or farm production practices, including animal welfare practices (Lagerkvist and Hess 2011) and farmland preservation (Duke et al. 2021). There has been less focus on consumer demand for producer attributes. There are a few important exceptions. First is an evaluation of the Homegrown by Heroes label, a label that indicates that the farm is owned by a veteran (Soley et al. 2019). Here, the authors found increased willingness to pay for fish with the applied label amongst Kentucky and South Carolina consumers, highlighting some public support for veteran-owned farms. Second, Zhao et al. (2023) evaluated consumer demand for muscadine grape jelly labeled to identify female farmers and beginning farmers. The authors found consumers were not willing to pay a premium for the jelly to support either female or beginning farmers, highlighting that public support for these two groups is more limited.

Outside of agriculture, demand for diversity in business ownership has been evaluated more recently. Diversity in agriculture lags far behind other areas in the food system—for example, while less than 10% of US producers are SDFRs, 41% of US restaurant firms are owned by diverse owners (National Restaurant Association 2022). A few papers evaluated the impact of the rollout of identity labels on online business profiles for restaurants. Aneja et al. (2025) evaluated the impact of minority ownership information for restaurants using the rollout of Yelp's “Black-Owned” restaurant labels, finding the labels increased restaurants' online traffic, calls, orders, and in-person visits. Similarly, Babar et al. (2023) evaluated the impact of

identity labeling using the online platforms, both through an experimental evaluation of demand for the labels applied to restaurants in Chicago, Illinois, and a broader evaluation of foot traffic for restaurants in Boston, Chicago, and Memphis with the label. They find that the label had positive effects for Black-owned businesses. More closely related, Moss et al. (2025) evaluated consumer preferences for Black Owned labels on three food products: BBQ sauce, beef jerky, and honey. The authors find that Black-Owned labels increased willingness to pay. There have also been some evaluations that have shown demand for labels such as “family farm” increase willingness to pay for some products (Wolf et al. 2011).

It is important to note that a particular label will not be utility enhancing for all consumers; rather, labels allow for appropriate matching between products of varying attributes and consumers with differing preferences (Caswell and Anders 2011). Previous research has similarly found that not all consumers prefer labels focused on production practices (e.g., organic) or farm ownership (e.g., local) (Nurse Rainbolt et al. 2012; Van Loo et al. 2011; Wolf et al. 2011). The literature on geographic labels has shown that demand is affected by the feelings of in-group members. For example, premiums for country, state, or region-specific food labels tend to be most preferred by consumers with ties to the area (e.g., Aizaki et al. 2023; Neil et al. 2020; Thøgersen 2023). For identity labels, current research shows less clear evidence of in-group/out-group effects. Soley, Hu, and Vassalos find that while consumer heterogeneity exists for veteran-owned farm labels (e.g., older consumers and consumers with higher incomes had higher premiums for the label), demand across consumers with and without connections to the military did not differ (2019). For Black-Owned labels, Aneja et al. (2025) find effects to be strongest in liberal areas; Babar et al. (2023) find higher demand amongst liberal and Black consumer groups; and Moss et al. (2025) find differences across gender, race, political party, and other consumer attributes.

The evaluations of Black-Owned labels in processed food products and restaurant websites provide evidence for public demand for increasing diverse business ownership. Yet, consumer demand for restaurant attributes and processed product attributes likely differ substantially from the demand for a label on a more commodity-like agricultural product. Additionally, the current literature is limited in both geographic and temporal scope. Currently, the majority of the literature has focused on large urban areas following Black Lives Matter protests, both of which are likely to have affected consumer responses. Here, using a national sample, we evaluate consumer demand for diverse owner attributes in agricultural products, and evaluate demand nearly 4 years after the start of the Black Lives Matter protests. Additionally, while all current evaluations of demand for diverse business ownership information have been more narrowly focused on a particular identity (e.g., Black Owned), we assess public demand for increased racial and ethnic diversity in agriculture more broadly, using a combined label for SDFRs. As we hold quality aspects of the product constant experimentally, we expect the changes in utility associated with the label to capture changes in consumers' expressive utility, meaning that consumers with positive utility for the label indicate demand for increased diversity in the food system.

3 | Design and Data

3.1 | Experimental Design

To evaluate consumer demand for increased diversity in agriculture, we utilized an online discrete choice experiment. For feasibility, we evaluated preferences using a single agricultural product: strawberries. We utilized strawberries due to the notable representation of SDFRs in the production of specialty crops and the product's wide consumption across the country (Yeh et al. 2023).⁴

We experimentally varied three relevant attributes: the USDA diverse farmer label, the USDA Organic label, and price (see Table 1). The diverse farmer label is a hypothetical label that identifies whether the product was grown on a farm owned by a Black or African American, American Indian or Alaska Native, Hispanic or Latino, or Asian or Pacific Islander farmer. All attributes were defined for all participants prior to answering the choice tasks.

For each choice task, participants were asked to choose between two one-pound boxes of strawberries with different labels and prices and an opt-out option. The range of appropriate prices was identified using 2023 USDA retail prices for organic and conventional strawberries in different regions of the US. Participants were asked to make 16 choices generated through a d-efficient design. The order of choice tasks and alternatives within choice tasks was randomized to prevent ordering effects. For each choice, the respondent was shown two images that had some combination of the organic label, diverse farmer label, or no label at different prices. An example of an alternative with both the organic and the diverse farmer label is shown in Figure 1. The full design is available in the Online Appendix A1.

Additionally, to test whether information on current diversity levels in agriculture would impact consumer preferences for a diverse farmer label, we randomly assigned half of the participants to see an information treatment before completing their choice tasks.⁵ Information treatments have effectively been used to evaluate marketing information in a variety of food choice contexts, including environmental information (Lin and Nayga 2022), descriptions of novel feed used in animal agriculture (Altmann et al. 2022), and explanations of genetic

TABLE 1 | Discrete choice experiment attributes and levels.

Attribute	Levels
Price	\$2.81, \$3.80, \$4.79, or \$5.78
USDA Organic	Claim or no claim
	
USDA diverse farmer	Claim or no claim
	

Note: The range of appropriate prices was identified using 2023 USDA retail prices for organic and conventional strawberries in different regions of the US.



FIGURE 1 | Example of an alternative.

production technologies (Hu et al. 2022). After reading the information, participants were asked to complete a knowledge check. Those who failed the check twice were excluded from the survey.

Although this work extends our understanding of consumer demand for diversity in agriculture, as with all hypothetical experiments, we are limited by hypothetical bias (e.g., List and Gallet 2001). One concern with hypothetical experiments is that when consumers do not spend their own money, they can present biased (often higher) willingness to pay for the product or attributes. To mitigate this bias as best as possible, we utilized a Cheap Talk script, which has been shown to reduce valuations in hypothetical experiments (Cummings and Taylor 1999).

3.2 | Survey Design

Before beginning the survey, respondents provided consent indicating that they would like to participate in the study. To be eligible, participants had to reside in the US and be at least 18 years old. Next, respondents answered demographic questions that included race, ethnicity, farming background, and political ideology. These questions were asked prior to the experiment to assure accurate responses. Third, participants read Cheap Talk text to reduce hypothetical bias. Fourth, participants completed a practice question to ensure that those using mobile devices could view alternatives simultaneously. Next, respondents are randomly assigned to the treatment or control group. Those assigned to the treatment read the text highlighting the current demographic makeup of farming in the US, and those assigned to the control read no text. Both groups then answered the same 16 choice tasks. Sixth, participants were debriefed about the design of the experiment, including the hypothetical nature of the diverse farmer label. Finally, respondents answered additional

questions about their preferences for diversity in agriculture. This survey was approved by the University of Illinois Urbana Champaign Institutional Review Board (#IRB23-0323). The complete survey instrument is included in the Appendix A2.

3.3 | Data

The survey was conducted online in February 2024. 1,527 participants were recruited via Qualtrics Panels using quotas to match the US population in terms of gender, age, annual household income, and census region, which is important as demand for food labels often differs across demographic characteristics. Sample characteristics are presented in column 1 of Table 2. Online surveys are not random samples and are limited by sample bias. While our sample is generally well-aligned with the characteristics of the US population, compared to the most recent US Census estimates, our sample underrepresents older consumers (75 years and older), Hispanic/Latino consumers, and the highest income consumers (\$150,000 and more). Additionally, compared to other estimates of consumer political ideologies, our sample also seems to overrepresent liberal consumers. For example, Gallup's 2023 estimates indicate that 25% of US adults consider themselves liberal, whereas approximately 30% of our sample indicated they are liberal (Jones 2024). Our sample has 757 participants who were randomly assigned to the treatment group and 770 participants who were randomly assigned to the control group. Participant characteristics were quite similar across the two groups (see Table 2, columns 2 and 3). Data is available in the supplemental materials.

4 | Methods

We estimate the demand for diversity in agriculture using a random utility framework (McFadden 1973). In this framework,

TABLE 2 | Sample characteristics.

Characteristic	Total sample	Control group	Treated group
Sex			
Male	47.7%	46.8%	48.8%
Female	52.3%	53.3%	51.3%
Age (in years)			
18–24	8.8%	9.9%	7.7%
25–34	20.6%	20.1%	21.0%
35–44	16.6%	15.5%	17.8%
45–54	15.5%	16.0%	15.1%
55–64	15.1%	16.0%	14.3%
65–74	16.6%	16.1%	17.0%
75+	6.8%	6.5%	7.1%
Annual household income (in \$)			
Less than 25,000	14.3%	14.3%	14.4%
25,000–49,000	20.7%	20.1%	21.3%
50,000–74,000	20.3%	22.2%	18.4%
75,000–99,000	15.0%	14.7%	15.3%
100,000–124,000	11.8%	11.2%	12.4%
125,000–149,000	7.5%	7.8%	7.1%
150,000 or more	10.4%	9.7%	11.1%
Region			
Northeast	17.2%	16.9%	17.4%
Midwest	21.2%	22.1%	20.2%
South	38.2%	39.2%	37.3%
West	23.4%	21.8%	25.1%
Ethnicity			
Not Hispanic/Latino	87.9%	89.4%	86.4%
Hispanic/Latino	12.1%	10.7%	13.6%
Race			
White	75.4%	74.3%	76.6%
Black or African American	12.3%	13.4%	11.1%
American Indian or Alaska Native	0.8%	0.6%	0.9%
Asian	4.4%	4.7%	4.1%
Native Hawaiian or Other Pacific	0.2%	0.3%	0.1%
Multiple Races	3.1%	2.9%	3.3%
Other	3.9%	3.9%	3.8%

(Continues)

TABLE 2 | (Continued)

Characteristic	Total sample	Control group	Treated group
Political ideology			
Liberal	30.4%	29.1%	31.7%
Moderate	40.8%	39.9%	41.7%
Conservative	28.8%	31.0%	26.6%
Number of observations	1527	770	757

the utility consumer i receives from choosing alternative j in choice task t can be defined as $U_{ijt} = V_{ijt} + \epsilon_{ijt}$. Here, V_{ijt} is the systematic portion of the consumer's utility and ϵ_{ijt} is the random and unobserved portion. If we assume the error term is independent and identically distributed according to the type I extreme value, we can estimate the probability of consumer i choosing alternative j in choice task t by integrating over the values of ϵ_{ijt} :

$$\Pr(U_{ijt} > U_{ikt} \quad \forall k) = \frac{e^{V_{ijt}}}{\sum_{k=1}^J e^{V_{ikt}}} \quad (1)$$

For our model, we define the V_{ijt} as a function of the experiment's attributes (see Equation 2).

$$V_{ijt} = \alpha + \beta \text{Price}_{ijt} + \gamma_1 \text{DFL}_{ijt} + \gamma_2 \text{Organic}_{ijt} + \gamma_3 \text{DFL}_{ijt} \times \text{Organic}_{ijt} + \delta \text{DFL}_{ijt} \times \text{Treat}_i \quad (2)$$

The α coefficient represents the utility relative to the opt-out. The β represents the marginal utility of price. The γ_1 captures the marginal utility for a diverse farmer label (DFL) compared to no label, γ_2 captures the marginal utility for the organic label compared to no label, and γ_3 accounts for the complementarity or substitutability between the two labels. We interact Treat, which takes the value of 1 when the participant was randomly assigned to the treatment group, with the diverse farmer label to see the impact of the information treatment on demand for the label. Thus, the delta coefficient accounts for whether exposure to information about SDFRs influenced marginal utility for the diverse farmer label (δ).⁶

In line with the literature, we expect responses to differ across consumers. We expect preferences for the diverse farmer label and responses to the information treatment to differ across in-group/out-group characteristics (e.g., race, ethnicity, and family connections to agriculture) and across consumers' political ideology. Thus, we utilize a random parameter (mixed) logit model, which allows preferences to vary across participants and is robust to violating the independence of irrelevant alternatives (IIA) assumption. We assume that price is distributed according to a negative lognormal distribution and all other coefficients are distributed according to a standard normal distribution.

Following the literature, we estimate the model first in marginal utility space, which is generally a better fitting model, and then estimate in willingness to pay space directly to obtain willingness to pay estimates with the most reasonable distributions

(Train and Weeks 2005). We use 550 Modified Latin Hypercube Sampling (MLHS) draws, which have been shown to be more appropriate than Halton draws (Hess et al. 2006). We also generate individual willingness to pay estimates for the diverse farmer label to examine how consumer attributes correlate to willingness to pay. We regress each consumer's coefficient draw of $\hat{\gamma}_{il}$ on consumer characteristics such as race, political ideology, income, and age. Analysis was conducted using the Apollo package in R (Hess and Palma 2019). Data and code are available in supplemental materials.

To further explore heterogeneity, we also estimate a latent class model. Latent class models, often used to explore heterogeneity in discrete choice experiments, estimate utility functions for different consumer classes and can be useful in identifying how demand differs across consumer segments (e.g., Chakrabarti et al. 2019; Ellison et al. 2022; Peschel et al. 2016). The latent class model builds on the conditional logit model (Equation 1), allowing choice probability to differ across consumer classes:

$$\Pr(U_{ijtm} > U_{iktm} \quad \forall k) = \sum_{m=1}^M S_m \frac{e^{V_{ijtm}}}{\sum_{k=1}^J e^{V_{iktm}}} \quad (3)$$

where m is the class and S_m is the probability of class m . Following the literature, the number of classes ($M = 3$) was selected by considering AIC, BIC, and interpretability (Boxall and Adamowicz 2002). In addition to the effects highlighted in Equation (2), we include socio-demographic characteristics of interest to explain segment membership. To explain class membership, we include an in-group/out-group measure (ROE) as a dummy variable that takes the value of one if the participant is racially or ethnically diverse and zero if the participant is both white and not Hispanic. We also include another measure of in-group/out-group by including a dummy variable for whether the participant has personal or family connections to agriculture (farm), which takes the value of one if the participant had connections to agriculture and zero otherwise.⁷ Following the literature on demand for diversity in restaurant ownership, we also include measures of political ideology (liberal and conservative are both compared to the omitted category, moderate). Finally, we include dummy variables to account for gender (female, which takes the value of 1 if the participant is female and 0 otherwise); age (Age 1, which is the base category and takes the value of 1 if the participant is between 18 and 34 years old, Age 2, which takes the value of 1 if a participant was between 35 and 54 years old and Age 3, which takes the value of 1 if a participant was 55 years or older), household income (HHI 1, which is the base category and takes the value of 1 if the participant's household makes less than \$50 thousand, HHI 2, which takes the value of 1 when the participant's household makes between \$50 and \$99 thousand, and HHI 3, which takes the value of 1 when the participant's household makes over \$100 thousand), and education (education, which takes the value of 1 if the participant received a bachelor's degree and a 0 if the participant's highest level of education was less than a bachelor's degree). We estimate the latent class model using Stata's llogit2 package (Yoo 2020). To compare estimates across classes, we compute willingness to pay for the labels using the ratio of attributes

TABLE 3 | Conditional logit results.

Coefficients	Estimates
Price ($\hat{\beta}$)	−0.94*** (0.01)
Diverse farmer label ($\hat{\gamma}_1$)	0.16*** (0.03)
Organic label ($\hat{\gamma}_2$)	0.59*** (0.03)
Diverse farmer label × organic label ($\hat{\gamma}_3$)	−0.20*** (0.04)
Diverse farmer label × treatment ($\hat{\delta}$)	0.12*** (0.04)
Constant ($\hat{\alpha}$)	4.94*** (0.05)
N choices	24,432
N participants	1527
Adjusted R ²	0.3142
LL	−18406.726

Note: Standard errors in parentheses; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

and the negative price effect. Data and code are available in [Supporting Information](#).

5 | Results and Discussion

5.1 | Conditional Logit Model Results

Our conditional logit results are presented in Table 3. As expected, we find that demand slopes down ($\hat{\beta} < 0$) and consumers have a positive preference for strawberries compared to opting out ($\hat{\alpha} > 0$). In line with previous literature, we find consumers have a positive marginal utility for the organic label ($\hat{\gamma}_2 > 0$). We also find that consumers exhibit positive preferences for diversity in agriculture ($\hat{\gamma}_1 > 0$) and the effect becomes stronger when they are exposed to information about current levels of diversity in agriculture ($\hat{\delta} > 0$). Preferences are stronger for organic than for the diverse farmer label. Namely, the premium for the diverse farmer label (~17 cents) is a little over 25% of the size of the organic premium (~63 cents), and, interestingly, consumers view the organic and diversity labels as having a substitutional relationship ($\hat{\gamma}_3 < 0$).

As discussed, conditional logit models do not allow for heterogeneity, which we expect is unrealistic in this case. Thus, we proceed to two sets of results that allow us to explore heterogeneity more specifically. First, we present results from our random parameter logit model, which allows for individual heterogeneity, providing mean and standard deviation estimates for the attributes, including the diverse farmer label. With our random parameter logit estimate, we also provide some evidence for what consumer segments prefer or oppose diversity in agriculture

TABLE 4 | Results from random parameter logit model in marginal utility space.

	Mean	Std dev
Constant ($\hat{\alpha}_i$)	23.55*** (1.694)	20.60*** (1.909)
Price ^a ($\hat{\beta}_i$)	0.49*** (0.056)	1.39*** (0.081)
USDA diverse farmer label ($\hat{\gamma}_{1i}$)	0.57*** (0.116)	1.48*** (0.130)
USDA organic ($\hat{\gamma}_{2i}$)	1.30*** (0.139)	2.28*** (0.130)
USDA diverse farmer label × USDA organic ($\hat{\gamma}_{3i}$)	−0.30*** (0.072)	0.41 (0.722)
USDA diverse farmer label × treatment ($\hat{\delta}_i$)	0.13 (0.161)	0.99*** (0.256)
N choices	24,432	
N participants	1527	
Adjusted R ²	0.576	
LL	−11375.4	

Note: Robust standard errors in parentheses; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

^aPrice takes a negative lognormal distribution, thus the estimated parameter's $\text{Mean}_{\text{price}} = -\exp(\beta_{\text{price}} + (s_{\text{price}}^2/2)) = -4.30$ and

$\text{sd}_{\text{price}} = \text{mean}_{\text{price}} \times \exp(\beta_{\text{price}} + (s_{\text{price}}^2/2)) = 10.12$

by evaluating correlations between estimated individual willingness to pay for the diverse farmer label attribute and participants' sociodemographic factors. Last, we present estimates from our latent class model, which allows utility to vary across consumer classes, and provide evidence for how demand for the attributes, including diverse farmer and organic labels, differed across classes.

5.2 | Random Parameter Logit Model Results

Table 4 presents the results of the random parameter logit model in marginal utility space. The constant is positive, indicating the utility of choosing strawberries over opting out of a purchase. Price takes a negative lognormal distribution, ensuring that demand slopes down for all consumers; thus, the estimated parameter's mean is −4.30, and its standard deviation is 10.12.⁸ We find that the diverse farmer label was associated with increased marginal utility for the average consumer. By comparing the size of the mean effect to the standard deviation, we can see that there is considerable consumer-level heterogeneity for the diverse farmer label, where not all consumers receive positive utility from the label; some received disutility. We explore heterogeneous responses in detail below. In line with the literature, we find that the organic label was associated with increased marginal utility. Additionally, we find that the interaction between the diverse farmer label and the organic label is negative, indicating that they are considered substitutes by consumers. We find that the information treatment did not have an impact

TABLE 5 | Willingness to pay estimates.

	WTP mean	WTP std dev
USDA diverse farmer label	0.14** (0.057)	0.97*** (0.072)
USDA organic	0.67*** (0.062)	1.47*** (0.049)
USDA diverse farmer label × USDA organic	−0.09*** (0.033)	0.13 (0.105)
USDA diverse farmer label × treatment	0.11 (0.094)	0.21 (0.179)
N choices	24,432	
N participants	1527	
Adjusted R ²	0.542	
LL	−12284.42	

Note: Robust standard errors in parentheses; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$; Results are from the random parameter logit model estimated directly in willingness to pay space. Full estimates in supplemental material.

on consumers' preference for a diverse farmer label on average; however, the significant standard deviation shows there is consumer heterogeneity in response to the information.

Table 5 presents an abbreviated output from the willingness to pay space estimates, highlighting the mean and standard deviations for the diverse farmer label, the organic label, their interaction, and the information treatment effect. For organic, we find that this label was associated with an average premium of 67 cents/pound of strawberries. If we compare this to the median price of a selected alternative (\$2.85), we find this translates to a 23.5% premium. For the diverse farmer label, we find an average increased willingness to pay of about 14 cents/pound (or 4.9% premium) and considerable consumer heterogeneity, which we explore in more detail below. We find these labels are substitutable on average; thus, non-organic SDFRs would likely benefit most from the label. Finally, we find that the information treatment highlighting the current demographic makeup of agriculture did not influence willingness to pay for the diverse farmer label on average.

We find considerable evidence of consumer-level heterogeneity for the diverse farmer label in our random parameter logit model. To explore this further, we present results from the linear regression examining the correlation of consumer demographics with participants' individual willingness to pay estimates for the diverse farmer label specifically (see Table 6).

We find that the average consumer had a positive willingness to pay for the label (\$0.153/lb). In line with previous investigations of responses to the restaurant label rollouts, we find that liberal consumers responded more positively (Babar et al. 2023; Aneja et al. 2025; Moss et al. 2025). Here, we find that liberal consumers had a higher willingness to pay for the label than moderate or conservative consumers, offering to pay an additional \$0.183/lb premium. Like Moss et al. (2025),

TABLE 6 | Correlation between estimated willingness to pay for diverse farmer label ($\hat{\gamma}_{it}$) and consumer characteristics.

Political ideology (base: moderate)	
Liberal	0.183*** (0.046)
Conservative	−0.052 (0.047)
Race (base: White)	
Black	0.107* (0.062)
Asian	0.154 (0.095)
Other	0.052 (0.075)
Sex (base: female)	
Male	−0.102** (0.041)
Age (base: 18–24)	
Age: 25–34	0.100 (0.078)
Age: 35–44	0.126 (0.083)
Age: 45–54	0.091 (0.085)
Age: 55–64	0.022 (0.084)
Age: 65–74	0.011 (0.084)
Age 75+	0.103 (0.102)
Farm background	0.056 (0.061)
Region (base: Midwest)	
Northeast	−0.047 (0.062)
South	−0.035 (0.052)
West	−0.022 (0.059)
Education (base: less than 4-year college degree)	

(Continues)

TABLE 6 | (Continued)

Political ideology (base: moderate)	
4-Year college degree or more	−0.076* (0.042)
Income (base: 50 k–100 k)	
Income: less than 50 k	0.001 (0.047)
Income: more than 100 k	−0.019 (0.050)
Constant	0.153* (0.092)
Observations	1527
Adjusted R^2	0.020

Note: Standard errors in parentheses; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

we also find a positive correlation between willingness to pay and participants being female (\$0.102/lb). Finally, we find no correlations between willingness to pay for the label and age, farm background, region, or income.

5.3 | Latent Class Model Results

Finally, we review results from our latent class model (see Table 7). Our model includes three classes: Class 1, which accounts for 17.5% of the sample; Class 2, which accounts for 46.2% of the sample; and Class 3, which accounts for 36.2% of the sample. We discuss results for each class in detail below. To calculate percent changes in willingness to pay, we calculate the change from willingness to pay for an unlabeled box of strawberries to a box of strawberries (e.g., for a box with a diversity label) for each class.

Class 1 was willing to pay about 6% (95% CI [0.026, 0.101]) more for the organic label. However, Class 1 was not willing to pay a premium for the diverse farmer label. The information treatment increased preference for the diversity label. Finally, Class 1 viewed the two labels' relationship as substitutional, although this relationship was only marginally significant.

Class 2 was not willing to pay for either the organic or diversity label individually. This class viewed the relationship between the labels as more complementary. As with Class 1, treatment had a positive impact on participants in Class 2.

Class 3 was willing to pay a 44% (95% CI [0.377, 0.498]) premium for the organic label and a 21% (95% CI [0.167, 0.253]) premium for the diversity label. Additionally, they viewed these labels as substitutional. Information also increased preferences for diversity, increasing the premium by 6% (95% CI [0.022, 0.105]).

The members of the classes differed across sociodemographic groups. Members of Class 1 were more likely to be male, older,

TABLE 7 | Latent class model results.

	Class 1	Class 2	Class 3
	17.5%	46.2%	36.2%
Choice attributes			
Price	−1.47*** (0.05)	−3.34*** (0.10)	−0.26*** (0.02)
Diversity label	−0.16 (0.10)	0.08 (0.12)	0.63*** (0.05)
Organic	0.31*** (0.09)	0.03 (0.10)	1.30*** (0.05)
Organic × diversity label	−0.22* (0.13)	0.63*** (0.16)	−0.33*** (0.06)
Treatment × diversity label	0.36*** (0.10)	0.28** (0.13)	0.19*** (0.06)
Constant	4.90*** (0.17)	17.24*** (0.49)	2.98*** (0.11)
Membership			
Female (base: male)	−0.38** (0.17)	−0.25* (0.13)	
35–54 years (base: 18–34 years)	0.18 (0.24)	0.03 (0.16)	
55+ years (base: 18–34 Years)	1.25*** (0.22)	0.55*** (0.16)	
HH income 50–99 K (base: < 50 K)	−0.31* (0.19)	0.16 (0.15)	
HH income 100 K+ (base: < 50 K)	−0.88*** (0.24)	0.15 (0.17)	
ROE (base: White, Non-Hispanic)	−0.17 (0.19)	−0.67*** (0.14)	
Education (base: less than bachelor's degree)	0.26 (0.17)	0.27** (0.13)	
Liberal (base: moderate)	0.07 (0.20)	0.08 (0.14)	
Conservative (base: moderate)	0.35* (0.19)	−0.07 (0.15)	
Farm family (base: no farm family)	1.03*** (0.32)	0.44** (0.18)	
Constant	−2.93*** (0.65)	−0.66* (0.37)	

Note: Standard errors in parentheses; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$;
 Characteristics that are associated with the likelihood to belong to Class 1 and
 Class 2 are compared to the omitted class, Class 3.

lower income, and have personal or familial connections to agriculture than those in Class 3 (the omitted class, see Table 7). Class 1 was also marginally more likely to be conservative than Class 3. Members of Class 2 were more likely to be male, older, white, and non-Hispanic, have received more education, and have personal or familial connections to agriculture than those in Class 3. This provides some evidence that those who are more likely to be supportive of diversity in agriculture (Class 3) are more likely to be female consumers, younger consumers, and those without personal or familial connections to agriculture. These results echo Moss et al. (2025), who find that demand for Black-Owned labels on food products was higher for female consumers and younger consumers. Unlike Moss et al. (2025), who find support for Black-Owned labels differed across race, we find that demand for a broader diverse farmer label did not differ across race.

5.4 | Measures of Stated Support for Diversity in Agriculture

Following the experiment, we asked participants an additional question about their level of support or opposition to the government (specifically the USDA) being involved in increasing diversity in agriculture. We find that 51.1% of participants were supportive, 29.4% were indifferent, and 19.5% were opposed. As with demand for a diverse farmer label, support was heterogeneous. Across politics, 70.9% of liberal consumers, 50.2% of moderate consumers, and 31.4% of conservative consumers indicated support. Across race and ethnicity, 47.1% of white, non-Hispanic consumers and 60.0% of non-white or Hispanic consumers indicated support. Additionally, 46.2% of male consumers and 55.5% of female consumers indicated support. Finally, 49.7% of those with personal or family connections to agriculture and 51.3% of those without connections to agriculture indicated support.

6 | Policy Implications and Discussion

While the current debate about diversity, equity, and inclusion continues at both federal and local levels, SDFRs face outsized challenges, and little is currently known about public support for diversity in agriculture. In this paper, we find evidence of public demand for increased diversity in agricultural ownership. In line with evidence from restaurants and processed products, our results show that support was heterogeneous, with some consumer segments (e.g., liberal, female) displaying higher demand for diverse farm ownership than other groups. Additionally, we find that when some consumers were informed about current levels of diversity in agriculture, preference for a diverse farmer label increased, highlighting the potential gap in knowledge among the public. Finally, our results show that 51.1% of participants indicated they support government involvement in increasing diversity in agriculture and support was heterogeneous across groups.

Together, these results provide useful insights about public demand for increasing diverse farm ownership for policymakers and other stakeholders, including grocery retailers weighing the continuation of their diverse supplier programs, farming

organizations identifying ways to best support SDFR members, and others. For SDFRs with well-aligned customer bases and production practices, these results show a potentially useful way to increase revenue. While our results show that stimulating consumer demand for diverse farmers through a label could be a complementary approach to other policy and market mechanisms aimed at increasing the profitability of SDFRs, the feasibility of such an agricultural labeling program is complex. Below we consider several potential issues.

First, for a label to be useful for a farmer, benefits must exceed costs. Farmers utilize labels to help differentiate their products and fetch higher prices; however, costs associated with labeling can also be high. For example, while organic premiums are often high, compliance with organic practices also increases costs substantially. For diverse farmer labels, the costs for qualifying farmers would likely be quite low. Comparable programs, including Homegrown Heroes and Women Owned Business, range in annual costs from \$0 to a few hundred dollars. Estimates above suggest that premiums would likely exceed these costs, particularly for non-organic farmers selling to customer segments that are well-aligned. Additional research should investigate demand for diversity labels in agriculture across a wider number of products, as, while we focus on strawberries here, it is possible that consumers would have different responses for different types of products (e.g., meat, dairy, other fruits/vegetables). Similarly, additional research is needed using revealed preference data to reduce concerns with hypothetical bias.

Second, to provide value, credence attribute labels must be believed by consumers willing to pay for the attribute. Credence attribute labels are often accompanied by a certifying body (e.g., government, NGO) that verifies the claims of the label and holds labelers to specific standards (Caswell and Anders 2011). While here we use the USDA as the hypothetical certifying organization (as it is likely to be known by participants), the agency is unlikely to be able to act as a certifier due to legal limits and unaligned current federal priorities. An NGO may be a more plausible certifier. For example, diverse owner claims for suppliers more broadly now often utilize third-party certification groups, including the National Minority Supplier Development Council, National Veteran Business Development Council, United States Hispanic Chamber of Commerce, and others to assure a business correctly indicates the characteristics of who owns, manages, and/or controls the majority of their company. For some platforms (e.g., Yelp), owners can self-identify as diverse without a certifying body. This would likely decrease cost and logistical hurdles, but may also decrease trust in, and thus value for, a label. Additional research is needed to evaluate public trust in such labels.

Third, the value of a diverse farmer label would differ substantially across farmers' individual sales strategies. For example, farmers who sell to retailers (including small and major grocers) or sell directly to consumers (including online sales, CSAs, and farmers markets) are likely more able to utilize such a label than those who sell into cooperatives or combined brands (e.g., Ocean Spray or Driscoll's). Additional research could evaluate eligible farmers' preferences and likelihood to use labels.

7 | Conclusion

In recent years, both government and industry have taken actions aimed at increasing the racial and ethnic diversity of participants in the food system (e.g., Johnson 2023; Mishra et al. 2024; Vekemans et al. 2024; Aneja et al. 2025) and in recent months, some of these actions have come under scrutiny (Holman and Creswell 2025; The White House 2025b); however, at this point, little is known about public demand for diversity in agriculture. In this paper, we aim to address this question using an online discrete choice experiment to evaluate consumer willingness to pay for a food label identifying SDFRs. Our findings provide evidence of public demand for diversity in farm ownership, with consumers willing to pay a positive price premium for a diverse farmer label. This demand, however, is not uniform across consumers, as we also find evidence of considerable heterogeneity.

Endnotes

¹ The USDA's SDFR definition included US farmers and ranchers who belong to groups that have been subject to racial or ethnic prejudice are known by the US Department of Agriculture. Specifically, the SDFR definition included Black or African American, American Indian or Alaska Native, Hispanic or Latino, and Asian or Pacific Islander farmers and ranchers. In some, but not all, USDA programs, this definition has been applied more broadly, including women, veterans, LGBTQ+, those with disabilities, and so on.

² At the time of this writing, many private sector programs meant to increase the number of diverse suppliers appear to have been maintained or criteria for applications broadened and identification labeling on both map and retail sites seem to have been generally maintained.

³ Caswell and Anders' labeling typology describes six types of labels across ownership of the labeling standard, means of certification, and labeling approach (voluntary/mandatory) (2011). Diversity labels in agriculture are most likely to be voluntary and either self-certified or third-party certified. As with many credence labels, firms often utilize third-party certification groups to increase traceability and believability. Current certifiers being used by diverse suppliers in retail include National Minority Supplier Development Council, National Veteran Business Development Council, United States Hispanic Chamber of Commerce, etc. Currently, some labels are utilized in retail more broadly—although have not been broadly applied in agriculture—to identify sellers' characteristics. Examples include Yelp's voluntary and self-certified "Latinx-Owned" restaurant label (Type II), Target's voluntary and third-party certified "Black-Owned" supplier labels, and the NGO-managed "Women Owned" label, which is a voluntary third-party label (Type IV).

⁴ Many agricultural products are sold under a combined brand label or co-op (e.g., Driscoll's, Ocean Spray), which may influence the manner in which identity labeling would be able to be utilized. For example, Women's Business Enterprise certification label requires those who utilize their label have 51% of their ownership be female. Additional research on how farm and brand ownership would be able or willing to utilize identity labeling is needed.

⁵ In our experiment, the treatment read: "In the US, farming currently lacks racial diversity—92.3% of all producers are white and not Hispanic. There has been increasing interest by the US Department of Agriculture in supporting farmers that have been subject to racial or ethnic prejudice—including Black/African American, American Indian or Alaska Native, Hispanic or Latino, and Asian or Pacific Islander farmers. These farmers have been historically excluded from land ownership, training programs, and federal support."

Participants also saw an accompanying pie chart, graphically showing the same racial and ethnic makeup of US farm ownership. The demographic information was drawn from USDA's 2017 Census of Agriculture (USDA 2019). The full survey can be found in the Appendix A2.

⁶We expected the information treatment to have an impact only on preferences for the diverse farmer label DFL, however, we also tested the interaction between the treatment and the organic label and the two labels' interaction, in case the information had an unexpected impact. Neither of these interactions were significant, and the inclusion of these interactions did not change estimates size, magnitude, or significance.

⁷The question asked participants, "Do you or anyone in your immediate family (grandparents, parents, siblings, aunts, or uncles) farm or ranch for a living?"

⁸ $\text{Mean}_{\text{price}} = -\exp\left(\beta_{\text{price}} + \left(\frac{s_{\text{price}}^2}{2}\right)\right) = -4.30$ and
 $\text{sd}_{\text{price}} = \text{mean}_{\text{price}} \times \exp\left(\beta_{\text{price}} + \left(\frac{s_{\text{price}}^2}{2}\right)\right) = 10.12.$

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Data S1:** Supporting Information. **Appendix S1:** Supporting Information. **Data S2:** Supporting Information.